

So, without any further delay, let's have a look at Anscombe's quartet.

The screenshot of the data set on Wikipedia is shown in Figure 1.

If you see the summary statistics of these four data sets, they are almost identical (to the last two decimal places at least). The summary statistics (for all the four data sets) are shown in Figure 2.

However, when you plot these four data sets, you will realise how different they are from each other.

To plot the data sets, follow the given steps:

Step 1: Download tables from the Wikipedia page on Anscombe's quartet

Step 2: Clean the data

Step 3: Plot the four data sets

Step 4: Plot the linear regression line for the four sub-plots

Step 5: End note

Step 1: Download tables from the Wikipedia page on Anscombe's quartets

We will download the data from the Wikipedia page using the `read_html()` function of a Panda's DataFrame. This function can download tables from an html page directly. Note the word 'tables' here (plural form). So, if a page has more than one table, it will download in the form of a 'list of DataFrame objects'. This Wiki page (on Anscombe's quartets) has, in fact, two tables. The first table gives the 'summary statistics' of the quartet, and the second table contains the four data sets.

The script to download the tables is as follows:

```
# Author:- Anurag Gupta ## E-mail 999.
anuraggupta@gmail.com
import pandas as pd
# ----- STEP 1:- Read the data set from
Wikipedia page
url = r'https://en.wikipedia.org/wiki/
Anscombe%27s_quartet'
df = pd.read_html(url)
# This wikipedia page has 2 tables.
# First table (index 0) has summary
```

Property	Value	Accuracy
Mean of x	9	exact
Sample variance of x : σ^2	11	exact
Mean of y	7.50	to 2 decimal places
Sample variance of y : σ^2	4.125	± 0.003
Correlation between x and y	0.816	to 3 decimal places
Linear regression line	$y = 3.00 + 0.500x$	to 2 and 3 decimal places, respectively
Coefficient of determination of the linear regression : R^2	0.67	to 2 decimal places

Figure 1: Data set on Wikipedia (Source: https://en.wikipedia.org/wiki/Anscombe%27s_quartet)

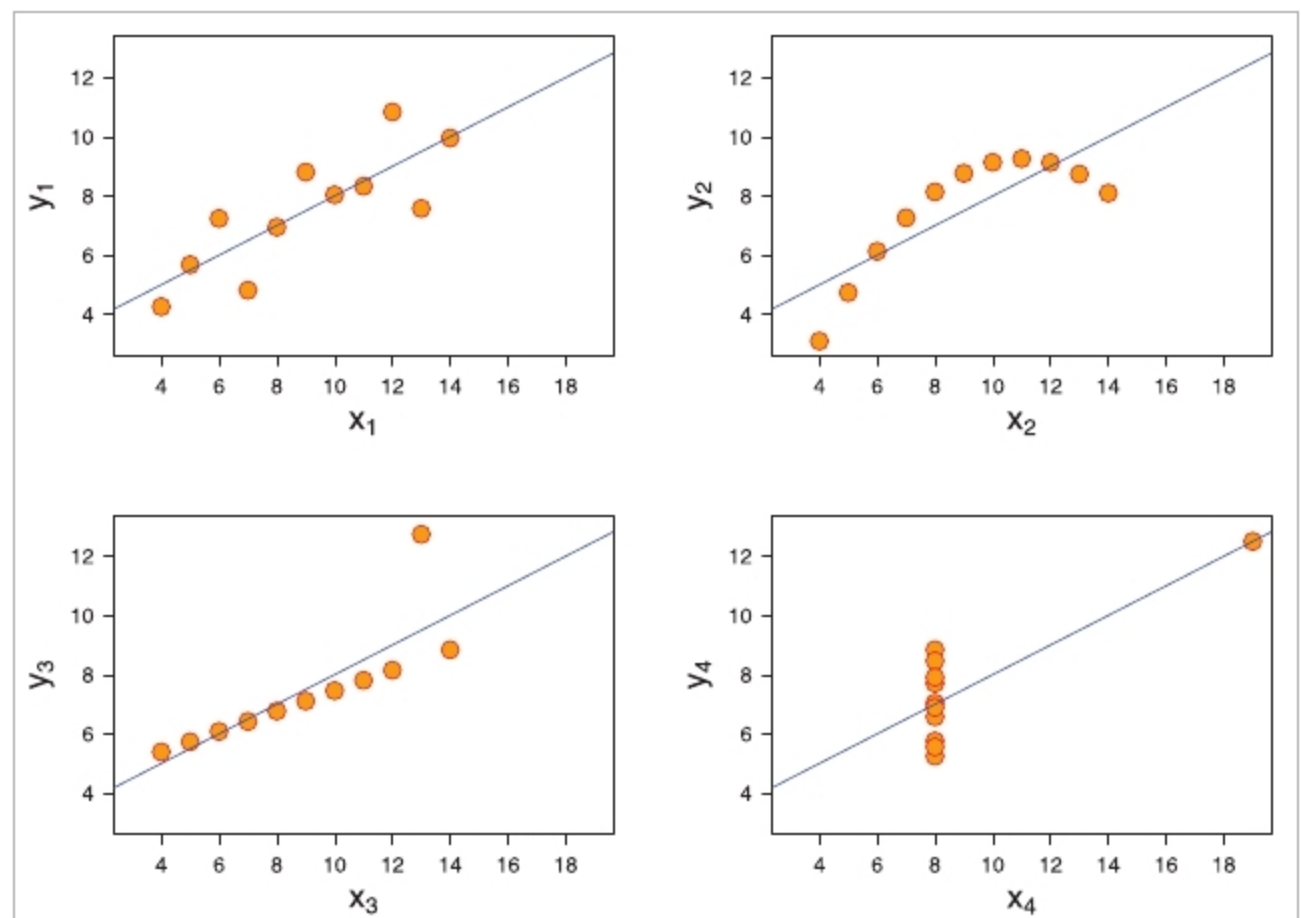


Figure 2: Summary statistics of the four data sets (Source: https://en.wikipedia.org/wiki/Anscombe%27s_quartet)

```
statistics
# Second table (index 1) has the 4 data
sets
print(df[0]) # prints summary
statistics of the data set
df2 = df[1] # df2 is simply an alias
for df[1]
print(df2) # Prints the 4 data sets in
from of a DataFrame
```

Step 2: Clean the data

We would like to point out an interesting thing about the data set downloaded from the Wikipedia page. Notice that in the screenshot of the four data sets (taken from the Wikipedia page), the column names are I, II, III, and IV, and there is one common column name for two columns. So

while there are eight columns, the column names are only I, II, III and IV. But Pandas is smart enough to recognise this and has named the columns as I, I.1, II, II.1, III, III.1, and IV, IV.1. Note that in some of the older versions of Pandas, the column names might show up as NaN. But that should not create a problem.

The cleaning of the data consists of the following three steps:

1. Rename the columns as X1, Y1, X2, Y2, X3, Y3, X4, and Y4.
2. Remove the row (at index 0) containing x and y. Note that in some older versions of Pandas, the row containing x and y may appear at index 1. This might happen if the Pandas takes the column names as